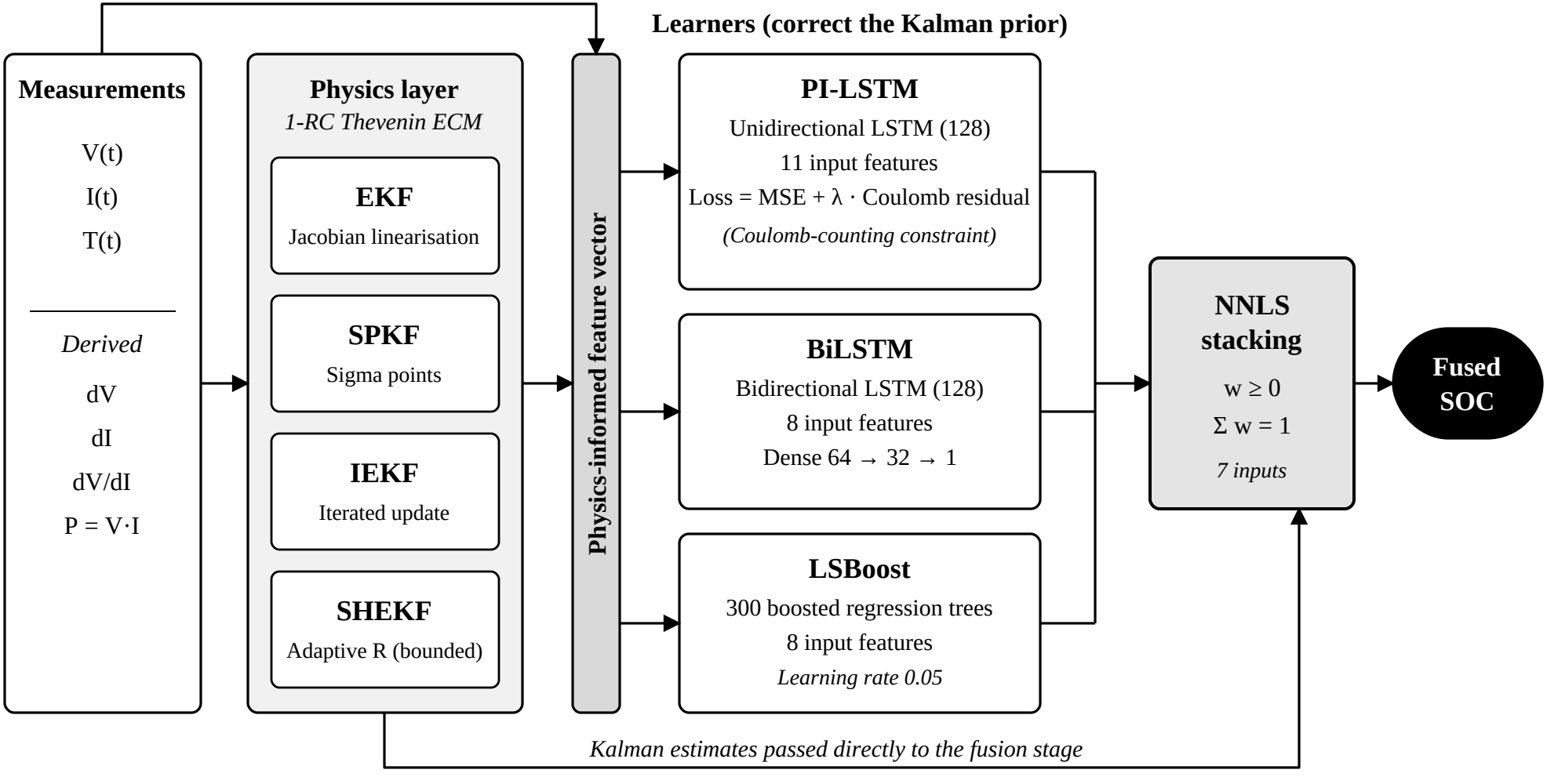
Learners (correct the Kalman prior)
Measurements
V(t)
I(t)
T(t)
Derived
dV
dI
dV/dI
P = V·I
Physics layer
1-RC Thevenin ECM
EKF
Jacobian linearisation
SPKF
Sigma points
IEKF
Iterated update
SHEKF
Adaptive R (bounded)
Physics-informed feature vector
PI-LSTM
Unidirectional LSTM (128)
11 input features
Loss = MSE + λ · Coulomb residual
(Coulomb-counting constraint)
BiLSTM
Bidirectional LSTM (128)
8 input features
Dense 64 → 32 → 1
LSBoost
300 boosted regression trees
8 input features
Learning rate 0.05
NNLS stacking
w ≥ 0
Σ w = 1
7 inputs
Fused SOC
Kalman estimates passed directly to the fusion stage